

Generative AI Notes

1. Introduction to Generative AI

Generative AI is a branch of Artificial Intelligence that creates new content by learning patterns from existing data. It can generate text, images, code, audio, videos, summaries, designs, and ideas from natural language prompts.

Core Pillars: • Learning from large datasets

- Understanding prompts and context
- Generating new outputs
- Improving through iteration and feedback

2. AI Model Training Process

Step 1: Collect and Prepare Data

- Gather data from text, images, audio, and code.
- Clean and organize data.

Step 2: Train the Model

- Learn patterns and relationships from data.
- Improve through repeated training.

Step 3: Generate New Outputs

- Accept new inputs and create original outputs.

3. Working of Generative Models

Input Data → Encoder → Latent Space → Decoder → Output

- Encoder compresses data.
- Latent Space stores important features.
- Decoder reconstructs or generates new content.

4. Generative vs Discriminative Models

Generative Models: • Learn data distribution.

- Generate new samples.
- Examples: GANs, VAEs, Autoencoders.

Discriminative Models: • Learn class boundaries.

- Predict labels and categories.
- Examples: Logistic Regression, SVM, Decision Trees.

5. Discriminative Models

- Used for classification and prediction.
- Learn decision boundaries between classes.
- Applications: Spam Detection, Image Classification, Trend Prediction.

6. Generative Models

- Learn hidden patterns from datasets.
- Generate text, images, audio, and videos.
- Create new data instead of only classifying.
- Examples: ChatGPT, DALL·E, Midjourney, Stable Diffusion.

7. Quick Revision

Generative AI = Creates new content.

Discriminative AI = Classifies or predicts outcomes.

Training Flow: Data Collection → Training → Pattern Learning → Content Generation.